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LearnNova:Automated Generation of Educational Content Using LLMs

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Author's Declaration

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Abstract

Many students struggle to understand mathematicsl and physics concepts when they study only through traditional text-based learning resources. LearnNova presents an automated system that helps to overcome this challenge. It generates educational video tutorials by using large language models together with animation and speech synthesis technologies. The system takes textual descriptions and turns them into Manim-based visual explanations. These visuals are then synchronized with automatically generated narrations. Through this process, the system produces clear and conceptually accurate videos that are easy to understand and memorize. It does not require learners or educators to have advanced programming knowledge. LearnNova aims to make high-quality visual learning accessible for both students and teachers. This approach supports better conceptual understanding through the use of intelligent automation.

Executive Summary

Learn Nova is a pioneering project designed to transform the educational landscape for STEM (Science, Technology, Engineering, and Mathematics) subjects. Traditional learning resources, such as static textbooks and lecture slides, often fail to adequately convey complex, abstract concepts like electromagnetic fields or vector transformations. Furthermore, the creation of high-quality visual aids currently requires specialized skills in graphic design or programming (e.g., Python), creating a significant barrier for educators and students alike.

This report details the development and validation of **Learn Nova**, an automated system that leverages Large Language Models (LLMs) to generate code for the Manim animation engine. This allows users to input simple text prompts—such as "Visualize a pendulum's motion"—and receive dynamic, mathematically accurate video tutorials instantly.

To validate the system's necessity, a survey was conducted with **70 respondents** from FAST-NU, comprising students and faculty. The results were compelling:

- **58%** of participants explicitly validated the need for such a system.
- **64%** identified "Difficulty Visualizing Abstract Concepts" as their primary learning hurdle.
- **50%** cited a "Lack of Technical Skills" as the main reason they do not create their own visual aids.

The findings confirm that Learn Nova fills a critical gap in the market. By democratizing access to high-quality educational visualizations, the system not only aids individual learning but also empowers educators to enhance their teaching materials without the overhead of learning complex animation software. The project concludes that AI-driven visualization is a viable and highly demanded supplement to traditional human-led instruction.

Table of Contents

List of Figures vii

List of Tables viii

1 Introduction 1

1.1 Introduction . . . 1

1.2 Problem Statement . . . 1

1.3 Objectives . . . 1

1.4 Scope . . . 2

1.5 Summary . . . 2

2 Project Vision 4

2.1 Problem Domain Overview . . . 4

2.2 Problem Elaboration . . . 5

2.3 Goals and Objectives . . . 5

2.4 Project Scope . . . 5

2.5 Business Opportunity . . . 6

2.6 Stakeholders Description / User Characteristics . . . 6

2.6.1 Stakeholders Summary . . . 7

2.6.2 Key High-Level Goals and Problems of Stakeholders . . . 7

2.7 Conclusion . . . 7

3 Literature Review / Related Work 8

3.1 Definitions, Acronyms, and Abbreviations . . . 8

3.2 Detailed Literature Review . . . 9

3.2.1 Video Generation . . . 9

3.2.2 Manim Code Generation . . . 17

3.3 Literature Review Summary Table . . . 26

3.4 Conclusion . . . 28

4 Results and Analysis **30**
4.1 Overview of the Survey . . . 30
4.2 Quantitative Analysis . . . 30
4.2.1 Demand for Automated Video Generation . . . 30
4.2.2 Barriers to Content Creation . . . 31
4.3 Qualitative Analysis . . . 31
4.3.1 Visualization of Abstract Concepts . . . 31
4.3.2 Dependency on Human Tutors . . . 32
4.3.3 Accessibility for Educators . . . 32
4.4 Discussion . . . 32

5 Findings **33**
5.1 Overview of Data Collection . . . 33
5.2 Quantitative Findings . . . 33
5.2.1 Demand for the System . . . 33
5.2.2 Identified Pain Points . . . 34
5.3 Qualitative Analysis . . . 34
5.4 Summary of Findings . . . 34

6 Conclusion and Future Work **36**
6.1 Conclusion . . . 36
6.2 Future Work . . . 36

List of Figures

4.1 Distribution of Respondent Agreement ($N = 70$)**.** 30

5.1 Survey results showing the demand for the Learn Nova system. 33

5.2 Frequency of major pain points reported by respondents. 34

List of Tables

3.1 Summary of Related Work for LearnNova 26

Chapter 1 Introduction

1.1 Introduction

Education in STEM (Science, Technology, Engineering, and Mathematics) fields heavily relies on the ability to visualize abstract concepts. From the behavior of subatomic particles in Physics to the transformation of matrices in Linear Algebra, understanding the dynamic nature of these systems is crucial for student success. However, traditional educational resources—primarily textbooks and static lecture slides—often struggle to convey motion and interactivity. While video tutorials exist, they are static artifacts that cannot be customized by the learner.

Learn Nova introduces a paradigm shift by automating the creation of these visual aids. By integrating the natural language processing capabilities of Large Language Models (LLMs) with the mathematical precision of the Manim (Mathematical Animation) engine, the system allows users to generate custom, high-quality video explanations simply by typing a text description.

1.2 Problem Statement

The core problem addressing this project is twofold, affecting both learners and educators:

1. **The Visualization Gap:** Students frequently encounter abstract concepts that are difficult to grasp through static media. The cognitive load required to mentally simulate complex systems (e.g., 3D vector fields) often hinders deep understanding.
2. **The Creation Barrier:** While animated videos are an effective solution, creating them requires significant technical expertise. Educators and students often lack the skills in professional graphic design software (like Adobe After Effects) or programming libraries (like Python/Manim) needed to produce their own visualizations.

Existing solutions are either too complex for the average user or too generic to address specific questions. There is currently no widely accessible tool that bridges the gap between a natural language question and a dynamic visual answer.

1.3 Objectives

The primary objective of Learn Nova is to democratize access to high-quality educational content. The specific goals of the project are:

- To develop an automated pipeline that translates natural language text prompts into executable

Manim code.

- To eliminate the need for technical skills (programming or graphic design) in the video creation process.
- To provide mathematically accurate and visually clear animations for core Physics and Mathematics topics.
- To validate the system's effectiveness and demand through a targeted survey of the academic community.

1.4 Scope

The scope of *Learn Nova* is defined as follows:

- **Domain:** The system focuses specifically on Mathematics (Calculus, Linear Algebra) and Physics (Mechanics, Electromagnetism), where the need for visualization is most acute.
- **Output:** The system generates short, silent or narrated MP4 video files focusing on a single concept per prompt.
- **Target Audience:** The primary users are high school and undergraduate students seeking to understand concepts, and educators seeking to create supplementary lecture materials.
- **Limitations:** The current iteration processes English language prompts and supports 2D animations via the standard Manim engine.

1.5 Summary

In summary, this chapter has highlighted the critical need for automated visualization tools to enhance STEM education. By addressing the limitations of traditional static resources and the technical barriers of manual video creation, *Learn Nova* proposes a novel, AI-driven solution. The defined objectives and scope establish a clear roadmap for developing a system that empowers both students and educators. With the foundational context and problem definition established, the subsequent chapters will detail the system's architecture and the findings from our validation survey.

The rest of the report is divided into the following sections:

- **Chapter 2 – Project Vision:** Defines the problem domain, project objectives, scope, and associated research context.
- **Chapter 3 – Literature Review:** Summarizes prior work and technologies related to automated

creation of educational content.

- **Chapter 4 –Results and Analysis** Discusses the results we get form the implemented methodology and analysis we made upon it.
- **Chapter 5 – Findings:** Describes findings on the basis of qualitative and quantitative analysis.
- **Chapter 6 – Conclusion and Future Work:** Summarizes the article, offers contributions, and potential improvements for the future.

Chapter 2 Project Vision

The LearnNova project envisions a system that is able to convert written educational content into high quality animated and narrated tutorial videos using Artificial Intelligence (AI) and Large Language Models; (LLMs). The idea behind it is to make tough concepts in mathematics and physics easy to grasp by automating the process of making such videos.Through this system, the project aims to link understanding and visualization by combining natural language processing , animation creation and speech synthesis technologies.

2.1 Problem Domain Overview

In education, one of the main problems is the lack of access to the material that can be used for interactive and visual learning particularly for abstract subjects like mathematics and physics. A large number of students still rely on static diagrams and textbooks, which do not always convey how complicated ideas or models actually function.

Modern visualization tools like **Manim** (Mathematical Animation Engine) helps to create clear, step-by-step animations for the users that make the abstract easy to follow. But the usage of such tools is not as simple as it seems to be. They require good programming skills, which most students and teachers lack.

LearnNova aims to cut this barrier by automating the process. The system accepts a written prompt or concept and converts it to a narrated educational video using the following modules:

- **Script Generation:** Using an LLM (for example, Gemini API) to write accurate and topic-based explanations.
- **Animation Generation:** Using a Manim tailored LLM trained with the 3Blue1Brown codebase to make sure everything is correct and can be visualized easily.
- **Text-to-Speech Narration:** Converting the generated script into natural narration by using a TTS system.

The final output is a high-quality tutorial video that helps learners to understand difficult concepts without any programming knowledge.

Problem Statement: *Students and educators lack an automated system that can transform conceptual text input into high-quality, animated, and narrated educational videos in mathematics and physics without requiring coding expertise.*

2.2 Problem Elaboration

Several existing systems try to generate videos automatically, but they still have major limitations when it comes to educational use:

- **Diffusion-based video models** (for example, Sora) can make realistic videos, but they do not maintain the step-by-step consistency that tutorials require.
- **Slide-based video tools** (for example, Google's NotebookLM Video Overview) can produce narrated summaries, but they lack the ability to show animated mathematical reasoning or transformations.
- **Prompt-based Manim tools** (for example, GenerativeManim and Manimator) depend heavily on detailed user input and often give mixed-quality results because they use general-purpose LLMs.

These issues demonstrate the need for a specialized solution. LearnNova contains a fine-tuned LLM trained specifically on high-quality Manim code from the 3Blue1Brown repository. This design means that the accuracy is ensured, the rendering is improved, and the explanation is clear and appropriate for learning. In combination with script generation, LearnNova also has modules for animation and narration, enabling the complete video creation process and not requiring any manual coding or design.

2.3 Goals and Objectives

The LearnNova project is driven by the following goals and objectives:

- To create an automated system for the creation of didactic video tutorials in mathematics and physics.
- To fine-tune an existing LLM using code from the Manim repository by 3Blue1Brown for accurate and reliable animation creation.
- To incorporate script generation, Manim-based animation, and TTS narration into a single workflow.
- To ensure that the animations produced are renderable and effective at teaching.
- To expedite the creation of visual learning material, make it scalable and accessible to all learners.

2.4 Project Scope

The key areas of the project are as follows:

- Focus on mathematics and physics topics only.
- Automatic script generation using an LLM API (for example Gemini)
- Animation generation through a fine-tuned LLM specialized in Manim.
- Use of the 3Blue1Brown version of Manim for consistency and correctness.
- Video duration between one and five minutes.
- A human-in-the-loop process for reviewing and refining the output.

The system will not include:

- Real-time or interactive editing features.
- Large-scale commercial use.
- Integration with external learning management systems or APIs.

The project aims to deliver a working prototype that shows a complete automated pipeline for creating educational videos.

2.5 Business Opportunity

The increasing demand for online and digital education opens up numerous opportunities for LearnNova.The system can be applied to e-learning sites, educational institutions, and tutoring centers. Some potential uses include:

- Producing personalised learning videos for individual students.
- Automating the production of content to be offered online in sites like Coursera or Khan Academy
- Providing visual content support for teachers and independent learners.

In the future, this technology can be packaged into a Software-as-a-Service (SaaS) product for the creation of educational content automatically.

2.6 Stakeholders Description / User Characteristics

LearnNova's success is based on the knowledge of the roles and expectations of the main users and stake-holders of the platform.

2.6.1 Stakeholders Summary

- **Students:** They are the primary users of LearnNova. The system will assist them in making animated video explanations for complex concepts and topics that they find challenging.
- **Educators:** Teachers and instructors can create visual resources for lectures and tutorials using LearnNova making lessons more interactive and comprehensible.
- **Developers/Researchers:** These are the team members who are improving the system by fine-tuning the model, developing the interface, and integrating the API.
- **Supervisors/Evaluators:** Academic supervisors and evaluators are responsible for overseeing the research process to ensure that the work meets the educational standard and that it complies with ethical standards of development.

2.6.2 Key High-Level Goals and Problems of Stakeholders

- **Students:** They need accessible, engaging, and visual learning tools that help them understand complex topics in STEM fields more clearly.
- **Educators:** They need accessible, engaging, and visual learning resources that help them better understand complex subjects in STEM fields.
- **Developers:**Their goal is to create a reliable and scalable system that automates the animation generation process with the help of AI.
- **Supervisors:**Their goal is to create a reliable and scalable system that automates the animation generation process with the help of AI.

2.7 Conclusion

In summary, Chapter 2 outlined the overall vision and strategic direction of the LearnNova project. It identified the existing gaps in educational visualization tools and explained how the proposed AI-driven framework aims to bridge those gaps through automated script generation, animation creation, and narration. The chapter also detailed the project's objectives, scope while highlighting potential business applications and key stakeholders. Collectively, these elements define the foundation upon which the system will be developed, guiding the technical and methodological work presented in the subsequent chapters.

Chapter 3 Literature Review / Related Work

This chapter presents the detailed literature review of studies relevant to video generation, manim code generation and training of domain specific LLM. A summary table is provided at the end to summarize the findings.

3.1 Definitions, Acronyms, and Abbreviations

To ensure a clear understanding of the terminology used throughout the chapter, a list of acronyms and their respective meanings are provided in the list below:

AI: Artificial Intelligence
API: Application Programming Interface
Diffusion Model: Probabilistic generative model that iteratively refines noise into structured data
DiT: Diffusion Transformer
FYP: Final Year Project
LLM: Large Language Model
LoRA: Low-Rank Adaptation (A parameter-efficient fine-tuning technique)
Manim: Mathematical Animation Engine
MLLM: Multimodal Large Language Model
NLP: Natural Language Processing
QLoRA: Quantized Low-Rank Adaptation (A parameter-efficient fine-tuning technique using 4-bit quantization)
RAG: Retrieval-Augmented Generation
SDG: Sustainable Development Goal
Transformer: Deep learning architecture based on self-attention
TTS: Text-to-Speech
VLM: Vision-Language Model

The provided list of abbreviations and definitions has been updated based on the content of the attached file, with new terms added and existing terms preserved, presented in alphabetical order.

3.2 Detailed Literature Review

This section offers a comprehensive review of studies that are relevant to our project. It details the techniques applied, the results obtained, and provides a critical analysis of these studies. The relevance of these studies to our project is also discussed.

3.2.1 Video Generation

Recent advancements in video generation have been largely driven by the evolution of diffusion and transformer-based generative models. Researchers have introduced frameworks capable of producing short, high-quality video clips directly from textual prompts, marking a significant leap in generative artificial intelligence.

3.2.1.1 AnimateDiff

Guo et al. presented AnimateDiff[1], which is one of the initial methods, enforced the diffusion-based image models onto animation in 2024. It accomplished this through the addition of motion modules in Stable Diffusion pipelines. This modular structure enabled regular short text animation with visual fidelity.

The authors in the article point to one of the fundamental weaknesses of the existing text-to-image diffusion models. The weakness is that there is no possibility to create animations and retain the visual quality and personalization of the base models. The article presents AnimateDiff[1] A plug-and-play framework is introduced in the paper. It can add motion functionality to any individualized text-to-image model, including Stable Diffusion, without the need to fine-tune the model. The system has a motion module, which learns general motion dynamics using video datasets in the real world. It also consists of domain adapter which guarantees stability of visual quality by reducing the gap between video and static image models. The authors, also, suggest MotionLoRA, a Light-weight fine-tuning method. MotionLoRA enables the model to acquire particular motion styles, including panning, zooming or rolling with little data and processing cost. AnimateDiff[1] is a modular architecture, using which personalized image models can be used to produce coherent, temporally continuous images and maintain their initial appearance and quality.

There are a number of common design philosophies when relating this work to LearnNova. The two systems focus on modular architectures that isolate different processes. In AnimateDiff animate in Diff [1], this is done by splitting motion learning and visual learning. In LearnNova, it is known as script generation, Manim code generation, audio narration, and video rendering. The concept of animateDiff is a plug-and-play motion module, which is aligned with the objective of LearnNova to create reusable elements of synchronized animation and narration. Moreover, the MotionLoRA fine-tuning method is similar to the strategy of the LearnNova to fine-tune general AI models to meet specific educational tasks. The fact that AnimateDiff[1] can be used to generate high-fidelity, controllable motion using the same model without retraining large models makes the vision of LearnNova automated generation of visually stimulating, correct, and synchronized educational videos cost-effective. Finally, although AnimateDiff[1] concentrates on general video creation using diffusion models, its concepts of motion modularity, fine-tuning efficiency, and domain preservation are direct inspirations of the LearnNova practice of intelligent educational content creation on a large scale.

3.2.1.2 VideoCrafter

Chen et al. based their work on these principles by coming up with VideoCrafter and its follow-up, VideoCrafter2 [2]. These models dealt with the shortage of datasets by using spatio-temporal adapters and multi-stage training. Their high-resolution generation was better in text-video alignment. These systems formed a fundamental open system of video synthesis systems that were developed in the future.

The authors of the article VideoCrafter2[2] describe the major issue of text-to-video generation. The question is how to produce movies that are aesthetic and flow well without having to employ very big and structured video collections. The artificial videos are generated by commercial programs like Runway Gen-2 and Pika Labs. These systems however rely on privately owned and vast video collections that are not accessible to the research fraternity. VideoCrafter2[2] attempts to address this issue by applying a strategy that trains diffusion-based video model using only low-quality video images and high-quality synthetic images. It is an aim to create superior videos without spending a lot of money on data. The approach segregates motion learning and appearance learning on the data level. Learners acquire motion through video with low quality like WebVid-10M. This is learned through high-quality synthetic images generated by models like Midjourney, to learn how appearance and picture quality. VideoCrafter2 [2] maintains the separation between these two types of learning to achieve good motion and clear visuals with the help of video resources of a large size.

VideoCrafter2[2] also examines the connection between space and time in video diffusion models, where such a model can be used to create videos that appear authentic and completely natural. It demonstrates that the process of learning spatial and temporal modules simultaneously makes a model stronger and more stable during fine-tuning. This assists the model to maintain motion even in significant cases of improvement in visual quality. The researchers employed the fine-tuning of LoRA to adapt the spatial modules with high-quality pictures. This technique enhances the realism of the image and maintains the movement of motion. The team evaluates their model on the benchmark of EvalCrafter. The findings demonstrate that the model is a good one compared to the best commercial systems. It achieves this in visual quality, correspondence between the text and the video and user preference, despite having significantly less data.

As we relate this work to the LearnNova project, we observe a lot of similar ideas. LearnNova is an AI based system which produces instructional video on subject areas like mathematics and physics. It does this automatically with large language models, Manim animations and text-to-speech tools. VideoCrafter2 [2] and LearnNova attempt to address the constraints in existing generative systems. Both apply fine-tuning, modular designs, and data efficient training. LearnNova is a problem in the teaching field. It should be able to produce the right, coordinated and straightforward educative videos without the technical expertise of a programmer. VideoCrafter2 [2] is concentrated on the uninterrupted movement and the clarity of the visuals in the overall video generation. In addition, two systems rely on LoRA or QLoRA fine-tuning to use the pre-trained models to fit specialized use.

3.2.1.3 Deforum

Throughout the development of the diffusion-based innovations, community-driven systems, like Deforum[3], came into existence. Deforum enables users to create animation-like videos with Stable Diffusion models based on an open-source framework. The system automatically synthesizes frame by frame images and uses these images to assemble them into an entire video. Users are able to make camera movements, keyframe based transitions and dynamic, prompt sequences without having to edit individually. Deforum finds one of the major obstacles of AI visual generation. It is generating smooth and time-coherent text-driven animations with creative flexibility. To control this, this framework gives numerous parameters. These are zoom, rotation, translation, interpolation schedules and keyframed attributes. These parameters provide users with the control over motion and change of scenes. The modular design enables video creators to incorporate text instructions, motion paths and rendering settings as an entirely automated generation process.

Deforum[3] defines how systems based on prompts can proceed through consecutive phases. The user will be required to first define prompts and animation settings. Afterward, the system creates latent images with the help of Stable diffusal. Every frame is drawn as per the set motion. Upon their preparation, they are then assembled into a final video file using tools like FFmpeg. The diffusion models are primarily fashioned towards artistic and stylistic images. Nevertheless, Deforum [3] features definite illustrations of the way in which the entire pipeline could handle the configuration, iterative creation, and output assembly. The project is about key framing, time based transitions and parameterized camera paths. It is possible to expressively and programmably animate behaviors with these features.

Comparing this work with the LearnNova project, a number of similarities can be observed. The two systems are similar in that they both work to automate the production of videos with the help of modular pipelines that convert textual instructions into full-blown videos. The frame generation and video compilation are not combined with prompt definition, which is separated by Deforum [3]. LearnNova isolates script generation, Manim code generation, TTS narration, animation rendering and audio-video syncing. The same way that Deforum[3] uses the animation settings as parameters that are based on keyframes, LearnNova synchronizes the timing of narration with the visual progression of mathematical scenes. It is demonstrated by Deforum [3] that it is possible to achieve the full automatization of video generation. This makes the objective of LearnNova to develop non-manual based educational animations more realistic. Deforum[3] however is concerned with the visuals of art as opposed to mathematical precision. This points to the necessity of a domain-specific system such as LearnNova, in which accuracy, learning pace and pedagogical clarity matter. Even with these variations, the two systems have fundamental concepts of automation, modularity and controlled video generation. Deforum [3] is thus a good source of information when developing scalable AI-based educational video pipelines.

3.2.1.4 CogVideo

Hong et al. later suggested the CogVideo [4]. It uses transformer encoders together with diffusion priors to enhance temporal reasoning and semantic alignment. The authors consider a major problem of text-to-video generation in CogVideo [4]. This problem is the time-varying understanding of complex actions and maintenance of visual quality despite the limited size of the training material or poor correspondence. As compared to text-image data, text-video data of high quality are scarce. Most of the datasets describe scenes and not movements. This complicates the production of videos according to the dynamic prompts such as a lion is drinking water by autoregressive models. CogVideo [4] proposes a simple solution to this issue by pretraining a 9.4-billion-parameter transformer model. It inherits spatial information of the text-to-image model CogView2. CogVideo, which inherently uses the spatial knowledge in images and concentrates on the temporal match, eliminates the necessity to train huge models in the first place and addresses the issue of poor text-video associations.

The article reports a number of technical advances in enhancing the quality of video and temporal consistency. Multi-frame-rate hierarchical training is the major contribution. This algorithm matches the text cues to the appropriate sections of a video by sampling the frames at various speeds. This makes sure that the model does not receive dissimilar fragments. CogVideo [4] divides keyframe generation and frame interpolation into two phases as well. This allows modeling of actions properly and retaining seamless transitions between frames. Dual-channel attention is another innovation. It enables the model to use the pretrained weights of spatial attention in CogView2 with an additional channel of temporal attention. This maintains the quality of the visual and enables the reasoning of time. The autoregressive generation is brought down to a slower pace by shifted window attention as it allows a part of the frame to generate in parallel. Using these components, CogVideo 0: Cogvideolargescalepretrainingtexttovideo is an excellent choice in machine and human-based evaluations. It performs better on visual clarity, temporal consistency and semantic accuracy over other open-source models that utilize text-to-video.

This article is connected to the LearnNova project in a number of aspects. The two systems resolve issues regarding the alignment of the various modalities with the time passage. CogVideo CogVideo [4] is based on learning temporal dynamics between text-prompt video frames and text prompts. LearnNova requires the accurate alignment of the narration schedule with the created Manim animation. Similar to CogVideo, which divides keyframe generation, and frame interpolation, LearnNova divides script creation, animation construction, audio synthesis, and video merging, in order to guarantee accuracy and coherence. The hierarchical training of CogVideo provided in the reference linked to Hong (2022): cogvideolargescalepretrainingtexttovideo) resembles the staged pipeline of LearnNova: each module enhances the output of the previous one. CogVideo[4] also uses a pretrained image model. This is like LearnNova that uses a narrow-tuned LLM to write Manim code without creating a code generator. Despite CogVideo being an ordinary video generation method, as well as LearnNova being an educational video generation method, both systems operate on the principle of modular processing, temporal consistency, and efficient model adaptation to generate high-quality videos automatically.

3.2.1.5 CogVideoX

The further CogVideoX model, which was proposed in 2025[5] proposed a large-scale text-to-video model. It seeks to enhance efficiency, quality, and access of video creation based on AI. Creating high-resolution, temporally consistent, and semantically accurate videos out of text in compressed fine-tuning and efficient diffusion architectures is the primary goal of CogVideoX[5]. It is intended to do a similar task as LearnNova that automates the creation of educational videos based on AI. They both seek to generate high-quality, context-sensitive, and synchronized visual data out of text.

CogVideoX [5] has a modular structure. It consists of a multimodal text encoder, a video diffusion model, and a video autoencoder to compress video representations making them efficient to store and fine-tune. Fine-tuning can be compressed to enable the model to learn new tasks and domains fast without compromising on the output quality. Its backbone is the Diffusion Transformer that guarantees long-range temporal continuity. This contributes to the smooth movement of motion and high visual content across frames.

CogVideoX [5] is a model that has been trained on large curated captioned datasets. It is doing well on benchmarks and it is even more realistic, the motion is more fluid, and semantically aligns with past text-to-video systems. Its findings indicate that diffusion-based generative models are able to generate coherent and high-resolution videos when fed text inputs.

Conceptually and architecturally, with regards to LearnNova, CogVideoX [5] offers some parallels. They both rely on text prompts in order to produce synchronized video. CogVideoX[5] is a system that creates movie scenes based on descriptive specifications. LearnNova creates learning animation both through script generation, Manim code rendering, and through synchronized narration. The two emphasize automation, accuracy and multimodal-integration to provide high quality output. The informal fine-tuning done by CogVideoX[5] instructs the direction taken by LearnNova to construct a self-hosted fine-tuned LLM to produce trustworthy Manim code. The two systems strive to minimize cost in computing and maintain scalability and precision.

Finally, CogVideoX [**?**] demonstrates how multimodal generation models can convert text into visually rich temporally coherent video. This is the driving force behind the vision of LearningNova that will allow it to automatize the process of video creation that relies on AI. They both combine text, animation, and audio, indicating the increase in the connection between the development of AI-based video generation and intelligent educational content creation. This opens the path to scalable, high-quality and accessible learning.

3.2.1.6 Wan 2.1

Wan 2.1[6] introduced by Alibaba DAMO Academy is another major initiative of this field. Wan2.1[6] Wan2.1 is a sophisticated video generation system. It is aimed at producing high-quality and temporally consistent videos using text prompts. It is dedicated to long-duration video stability and elaborate scene building and effective management of sophisticated visual sequences. In spite of the fact that the Wan2.1[6] is designed to take up general video synthesis jobs, part of its fundamental concepts can be used to inform our project improvements. In our project, we will automate the creation of learning videos which will contain narration and Manim-based mathematical animations.

The fact that Wan2.1[6] uses a single framework that enables it to generate text-to-video, taking into account the structure of the scene and its temporal flow, is one of its strengths. Our system can be applicable to this approach. In LearnNova, any given animation has to be in line with the flow of the narration and should be visually consistent throughout the video. We do not make use of Wan2.1 [6] per se. Nevertheless, the focus on synchronization and stability allows demonstrating the significance of timing and seamless transition between frames in our workflow.

Wan2.1 [6] also emphasises efficient generation pipelines which minimise the appearance of the visual artifacts in the longer sequences. Our project also relies on the generation of clean and artifact-free animations. Mathematical images should be clear and accurate. The concept of having world consistency in long video clips is helpful in the context of rendering multi-step visual explanations in Manim.

Lastly, the paper of Wan2.1[6], which is based on the concept that large video generation models can combine text comprehension and visualization within a single system, is interesting to observe how the two capabilities can be combined into a single model. These duties are distanced in our arrangement. The Gemini API script is used and the Manim code is written by a fined-tuned LLM. Although the architecture is different, Wan2.1 confirms the need to maintain the same meaning, pacing, and structure of both the script and visuals. This will help us achieve our objective of producing coherent video educational content wherein narration and animation complement each other.

3.2.1.7 HunyuanVideo

Equally, HunyuanVideo by Tencent[7] suggested a multimodal framework. It combines visual, textual and auditory content leading to the stable long-duration motion generation. The two works exhibit the move towards more general-purpose, high-fidelity video generators that are able to create more complex scenes. Nevertheless, they focus on the cinematic realism, as opposed to the educational visualization.

HunyuanVideo [7] is an open-source video foundation model. It is geared towards being as good or better than closed-source systems. It is a multimodal learning model that uses state-of-the-art data curation, large-scale diffusion navigator and semantically matching texts to generate realistic and semantically consistent videos with textual prompts.

HunyuanVideo [7] has a systemic pipeline. It has a full-fledged data filtering, organized annotation, and progressive pre-training based on images and videos. This model architecture combines a 3D Variational Autoencoder to reduce the size of visual data and a diffusion model based on Transformers to create dynamic video sequences. A Multimodal Large Language Model processes text prompts, ensuring that there is a high alignment between processing language and visuals.

The training in the system is based on progressive curriculum. It begins with low-quality clips and progresses to quality and long-lasting videos. This gives an equilibrium between efficiency and fidelity. HunyuanVideo[7], which has more than 13 billion parameters, provides a high quality of motion, text-video alignment, and realistic images. It performs better than models in the commercial market such as Runway Gen-3 and Luma 1.6. The model can generate context-rich and multimodal high-resolution videos with its structured captions, the ability to recognize camera movement, and multimodal integration.The framework has applications to several downstreams. These are video to audio, image to video conversion and avatars. It focuses on developing synchronized, expressive, and multimedia content that is immersive.

Regarding LearnNova, two systems have in common the purpose to automate the creation of content with the assistance of AI-based multimodal generation. HunyuanVideo [7] is devoted to video synthesis and understanding. LearnNova is involved in the generation of educational video. In LearnNova, the script generation with the Gemini API is automatically transformed into animation generation with fine-tuned Manim LLM and audio synchronization with TTS. These two structures are focused on precision, timeliness, and automation. They demonstrate that multimodal AI could be used to improve content creation and accessibility.

In this way, HunyuanVideo [7] offers solid conceptual and technical basis of LearnNova. It demonstrates the way the developed multimodal architectures, structured pipelines of data and text-video alignment methods can be modified to produce AI-based educational videos. These videos are pictorially consistent, semantically accurate, and didactical.

3.2.1.8 Community Driven Models

Further progress has been seen in open and community-driven models. HPCAI Tech's Open-Sora [8] democratized transformer–diffusion architectures through open-source implementations, focusing on efficiency and accessibility for wider research use. Lightricks Research developed LTX-Video [9], prioritizing real-time generation through a low-latency latent diffusion pipeline. These efforts highlight the growing interest in balancing performance and responsiveness, though they often compromise temporal consistency for speed. Other applied frameworks such as Mochi 1 by Genmo AI [10] explored prompt-driven video storytelling, offering narrative control through natural language input. Projects like SkyReels [11] and ComfyUI–VideoCrafter integrations [12] extended usability by focusing on user-centric interfaces and modular node-based workflows for non-expert creators.

3.2.1.9 Research Gap

Collectively, these advancements demonstrate remarkable progress in video generation, showcasing strong potential for automated visual content creation.The cited papers and models collectively represent the technological foundation and motivation for the LearnNova project. Works like AnimateDiff [1], VideoCrafter2 [2], and LTX-Video [9] demonstrate how diffusion-based architectures can generate short, high-quality videos but still struggle with temporal consistency—an essential feature for educational tutorials.CogVideo [4] and CogVideoX [5] introduce transformer-based text-to-video generation with stronger semantic understanding, showing potential for script-driven animation but prioritizing cinematic realism over pedagogical clarity. Similarly, HunyuanVideo [7] and Wan 2.1 [6] explore multimodal and large-scale video generation, aligning conceptually with LearnNova's integration of text, visuals, and audio, though they lack fine-grained control for concept explanation. Open and community-driven systems like Open-Sora [8], Deforum [3], ComfyUI–VideoCrafter [12], SkyReels [11], and Mochi 1 [10] emphasize accessibility, modularity, and creative freedom—principles central to LearnNova's vision of democratizing educational video creation for non-technical users. While these models focus on realism, storytelling, or general-purpose generation, LearnNova extends this progress toward domain-specific, pedagogically accurate, and automated Manim-based animation, filling a critical gap in AI-driven educational content generation.

The gap identified through this review lies in the absence of domain-specific systems that prioritize conceptual accuracy, structured narration, and pedagogical intent. LearnNova addresses this gap by integrating large language models for script generation, specialized fine-tuning for Manim-based animation, and synchronized text-to-speech narration. This approach combines automation, accessibility, and domain alignment to generate precise and visually coherent educational videos in mathematics and physics.

3.2.2 Manim Code Generation

This section have explored the automation of educational content creation through multimodal systems combining text, code, and video generation.

3.2.2.1 Theorem Explain Agent

Theorem Explain agent was introduced by Li et al[13] a video based multimodal explainations. It is an agentic system, where LLMs are used to create Manim based animations of theorems proving. Their method is symbolic and visual. It demonstrates how the abstract mathematical reasoning could be transformed by LLMs into animated descriptions that can be comprehended by people. Both visual reasoning and textual reasoning have been employed in the framework in this work. The research broadens the understanding of AI interpretability. It does it by combining big language models, agent planning, and automated visualization using Manim animation. The system attempts to relate symbolic reasoning and human knowledge. It achieves this by providing structured and multimodal explanations in the STEM.

TheoremExplainAgent [13] has two massive language model agents that collaborate. The first is a planner agent. It develops the plot and the storyboard. The second is a coding agent. It converts the plan into Manim scripts which are executable. This architecture is a reflection of notions of agent based AI systems. In such systems tasks are planned and done through the use of tools and also through interaction of the model with the environments. A retrieval augmented generation technique is also present in the study. In this approach, the agents are able to use documents, visual examples, and code templates when generating. This assists in the enhancement of animation. By combining these techniques, Theorem

Explain Agent[13] demonstrates that the systematic video creation is capable of explaining complicated theorems in an understandable manner.

Theorem Explain Agent [13] pays attention to multimodal theorem explanation. Our system applies this philosophy to the general educational video generation. It seeks to make 3Blue1Brown style mathematical animations in an automated system which involves the use of AI. In order to assess multimodal reasoning in a systematic manner, the authors introduce Theorem Explai nBench. It is a standard that consists of 240 theorems in mathematics, physics, chemistry and computer science. The theorems are categorized into various levels. The benchmark employs five metrics to evaluate the benchmark. These measures are accuracy and depth, visual relevance, logical flow, layout of elements and visual consistency. This design assists in the measure of factual quality and visual quality. It resembles prior multimodal reasoning systems including Science QA and Theorem QA. It is concerned with multimodal explanation and not text only reasoning.

LearnNova takes the form of the two agent system in Theorem Explain Agent[13]. It uses modular automation. The structured script generation is made with the help of the Gemini API. We also call a fine tuned LLM which generates Manim animation code of that script. This is according to the planner and coder concept in Theorem Explain Agent[13]. It transposes this concept to an educational situation that extends beyond the reasoning of theorems.

Both systems apply retrieval and reasoning systems in enhancing accuracy and coherence. Retrieval such as optimization occurs in our system when the LLM retrieves finely tuned Manim templates and patterns of animation. This is beneficial in making sure that generated visuals are correct in a teaching environment and sound in a mathematics sense. Our pipeline is based on a text to speech (TTS) integration that is detailed by the idea in TheoremExplainAgent. It combines text, graphic and story telling. It does so to allow conceptual clarity.

Our project has a conceptual underpinning by the Theorem Explain Agent[13]. It endorses the notion that Manim generation through LLM acting as the means of AI based teaching, learning and explainability. As we derive its multimodal reasoning and agentic planning concepts, our project will be remodeled to an automated structure. This model will deliver correct, coordinated and visually stimulating educational videos. It also facilitates the larger objective of AI assisted teaching and generation of content. The findings indicate that the working of Theorem Explain Agent[13] is more efficient with long form video generation compared to text to video models. Certain issues still abide in graphical presentation, consistency and retrieval of information. These are the goals that LearnNova attempts to achieve. The challenges can also be found in other works concerning retrieval augmented code generation.

3.2.2.2 Manimator

On the same note, Herrmann et al. [14] invented Manimator. It is a program transforming scientific text into Manim animations. It seeks to simplify the research conceptualization process. The system generated attractive findings. It lacked the synchronized narration. It also failed to provide precision in timing and learning pace.

Manimator[14] is an open source system. It applies Large Language Models to research papers and natural language descriptions to generate explanatory animations with the help of Manim engine. The primary objective of Manimator [14] is to simplify the process of grasping scientific and mathematical work. It achieves this through transforming the written forms into moving images. This can be compared with the use of simple diagrams to describe complicated concepts in a teaching process.

The process of manuscripting research papers in Manimator[14] has three stages. The initial step is concerned with the knowledge of the input and transformation into an organized visual plan. A large language model that is multimodal analyses the source. The source can be a research paper, textbook PDF or plain text prompt. In the case of complex inputs, visual and textual comprehension is done by models like Gemini 2.0 Flash. LLaMA 3.3 70B supports prompts consisting of text only. The model recognizes definitions, equations, theorems and conceptual metaphors. It organizes these items in a Markdown Scene Description file. This file includes the list of the topics, major LaTeX equations, visuals suggested, and the list of the animation flow. This step transforms elaborated academic content into a step by step storyboard. This storyboard is to be rendered into a visual later on.

The second stage starts when the structured scene description is prepared. This step translates the plan into Python code with the help of a code specialized LLM like DeepSeek V3 or Qwen2.5 Coder. This model is a specialist in Manim engine. It adheres to some examples to be consistent. It also produces classes, scenes, and transitions in order of sequence and timing. It also includes mathematical notations. It establishes fluid animations. It does not offer visual clutter as it does not overcrowd. This phase transforms the conceptual descriptions in Manim scripts that demonstrate the logical flow and the visual flow of the information.

The final step involves the Python script being run using the Manim library in order to produce an MP4 animation. Dependencies like FFmpeg regulate the compilation of the video and provide a fluent output. The ideas are clarified in the animation that has been completed with movement, text, and graphics. This gives abstract subjects a lot easier time to follow as we do when we give out lessons step by step. TheoremExplainBench benchmark was used to test the performance of Manimator. It achieved a similar system accuracy of 0.845. The evaluation of the human output was provided by the engineering students and the evaluation indicated that the output was clear and well organized. Animations were found to be useful by the students in order to comprehend academic work.

During this study it was revealed that production of Manim animation typically involves programming skills, as well as, background knowledge. The system was tested on the TheoremExplainBench benchmark. It was very accurate and was well responded to in terms of clarity and coherence. It has shown success by indicating that programming skills are not required in the generation of animation in education on an automatic basis. It is also able to simplify visual learning to learners.

It is the idea on which our project LearnNova is based. It has an extended goal. Manimator[14] lays attention on scholarly articles and theory. LearnNova is an automation system that is used to produce educational videos. It produces Manim animation. Synchronized narration and script generation are also part of it. In LearnNova, a user gives a topic prompt. The script is created on the Gemini API. Fine tuned LLM generates the animation code of Manim. The text is converted to audio through a text to speech engine. The audio and the animation are combined to a final educative video.

To conclude, the two systems attempt to make the learning process easier by using AI generated visualizations. LearnNova takes this vision one step further, creating the entire video tutorials that have narration and animations. It provides a more interactive automation in the development of educational content in mathematics and physics. The similarity between the two systems is the aim of the systems, that is, to make visual learning more accessible, scalable and effective using animations.

3.2.2.3 AutoManim

Simultaneously, an attempt to automatize Manim script generation with the help of natural language inputs was introduced in AutoManim[15]. It was a system which relied on pre trained language models to produce simple scripts. It had some success in syntactically correct code. Simultaneously, the semantic errors were frequent in the animations. There were also problems in teaching flow and clarity. By so doing, the findings demonstrated the initial weaknesses and the shortcomings of this technique.

A straightforward Python package was launched in the automanim [15] project. It was intended to generate Manim animation using a description in plain text. It was supported on Python 3.10 and beyond. Automanim [15] was intended as a simplification of making mathematical or conceptual animations. It attempted to minimize the manual Manim scripting. The package analyzed the text that a user inputted and converted it into a Manim scene or animation script. This initial release included few features because the first version was 0.0.1 released in June 2025. It concentrated on easy workflows to Manim animation.

The scope of the project was small. Meanwhile, its key concept provided a significant trend. The concept of natural language generation of Manim code was linked to broader efforts to simplify technical and educational types of visualizations. The project demonstrated the process of transformation of natural language to animation code. It was also used as an example of automatic code generation to mathematical animations.

This concept corresponded to the objectives of the LearnNova system. LearnNova was engaged in automation of educational video production. It relied on a pipeline that consisted of script generation, Manim code generation, text to speech narration, animation rendering and audio video synchronization. Whilst automanim paid attention to the transformation of text into Manim code alone, LearnNova extended the procedure. It introduced timing of the narration, careful rendering and numerous processing levels. Automanim in this manner served as a precursor to text driven animation generation. LearnNova was carrying on with this concept and created full-length and synchronized videos. The videos were made ready to be used in teaching and learning.

3.2.2.4 Text2Video

There has also been complementary research on end-to-end systems of automated video creation. Text2Video system

Text2Video study by Yu et al. [16] discusses how the increasing need to use multimedia production tools efficiently has led to the development of a system that enables the generation of game-related videos based on the text scripts automatically. Similarly to the step-by-step approach to cooking a dish with the help of a recipe, their system takes the form of a structured workflow which is a combination of natural language processing step, cross-modal retrieval, and automated video editing. This enables even the beginner users to create videos without professionalism.

Five principal modules are present in the architecture namely material management, NLP, cross-modal retrieval, TTS and video editing. Raw video, audio and images are collected and sorted out by the material manager. It considers important in-game moments (e.g., first blood' or triple kill' and so on) by using a highlight extractor with feature matching, OCR, and convolutional neural networks to identify heroes. The semantic labels are annotated on all materials, similar to the tagging of items in a library to allow easier search in the future.

In general, the Text2Video[16] system is a step towards the creation of automated multimedia, particularly, in terms of the gaming-related content. It serves as a bridge between text, description and end video production, and can allow users to develop rich, coherent visual content with minimal manual work. It enables new applications to AI-assisted storytelling and user-generated media by integrating state-of-the-art NLP, computer vision, and speech synthesis.

Likewise, same like the Text2Video[16] platform will be automated in terms of video generation using text scripts, in our project, we take this concept and apply it to educational and mathematical visualization.

In our suggested system, the processing of 3Blue1Brown style math videos is automated through scripting, code synthesis with an AI-based framework, and multi-media rendering. The system does not use the manual animation or a hand-written script, but it reads a topic prompt as a source of input and applies the Gemini API to create a script to deliver an educational message. Instead, a fine-tuned LLM, specifically trained to generate Manim code, is used, which then generates synchronized mathematical animation, which visually matches the concepts defined in the script.

The script is converted into natural speech sound by a text-to-speech module with a timely effect in order to facilitate a seamless learning experience. The images depicted in Manim and the audio created are blended into a final video that is synchronized. This eliminates the requirement of either manual coding or video editing expertise and the high-quality production of mathematical videos is more readily available to teachers and learners.

On the one hand, the system developed by Yu et al.[16] gave more emphasis on multimedia retrieval and game narration. On the other hand, our system is concerned with correctness, visualization, and education content. Our project incorporates code generation through the use of LLM, TTS alignment, and Manim animations, which can represent a new trend in AI-assisted educational media. It facilitates the development of factual, understandable, and aesthetical learning materials.

3.2.2.5 EduViz

Similarly, EduViz [17] embraced a new level of automated visualization in STEM education by integrating textual knowledge with interactive visualization. Nevertheless, the narration, changes of scenes, and the development of the concepts were not highly correlated. In Advance EDU-AR-VIZ: a Framework to Choose the Right Visual Augmentations in STEM Education [17] the authors attempt to make the abstract STEM concepts comprehensible to the learners. This is done through visual and interactive media. They bring out a framework of carefully selecting visual augmentation techniques. They emphasize an augmented-reality learning application on the mobile platform teaching magnetism. Their approach consists in the choice of visualizations which correspond to learning objectives, the prior knowledge of students, and the technology they have. They also create and test an app on Android mobile AR. In their findings, they reveal that even when the students are conversant with the subject, they find the lessons interesting and useful.

The article provides some of the most important aspects that can be utilized in the development of educational videos. To begin with, it is not enough to demonstrate images. Types of visual augmentations should be selected depending on the depth of the concept, requirements of students and the approach of interaction. Second, AR experiences on the phone can facilitate active learning. Students are provided with spatial and interactive management of things that might have remained invisible or abstract. Taking an example, magnetic field lines are possible to be depicted. Thirdly, one should align technology and teaching objectives. An example would be when to resort to three-dimensional visualizations and not the use of simple two-dimensional overlays. The results of the evaluation prove that students consider AR-supported lessons easier to follow and more motivating. This demonstrates that effective visual augmentations are useful in learning.

Comparing this to the LearnNova project, we can see a few similarities. The goal of LearnNova is also to produce high quality educational videos automatically. It involves synchronized narration and mathematical animation. The visual augmentations in this instance are developed with the code-driven Manim animations, rather than with AR. The outlay of the EduViz [17] demonstrates the reason behind choosing the appropriate visualization. This, in LearnNova, involves the right Manim animation style, pace and time of narration to fit what the learners already know and the learning objective. Like EduViz[17] was interested in aligning the visual design with the teaching objectives and the learners, LearnNova makes sure that the script and Manim work together with the audio and the end video. These two systems are concerned with modular design. EduViz[17] is based on a visual selection framework and LearnNova is based on a script, code, audio, and render pipeline. They both strive to make STEM education more inclusive and interesting.

3.2.2.6 Multi-RAG

Multi-RAG[18], a project recently developed, suggested a system to enhance the ability of AI to make sense of videos better. Multi-RAG is a video, audio, and text combination that is generated through Retrieval-Augmented Generation. The primary purpose is to enable AI to provide objective and contextual responses as opposed to assumptions.

There are two issues with the traditional large language models. To begin with, they do not have the access to the updated data. Second, they occasionally produce unreal or fantasized information. Multi-RAG[18] addresses these issues by searching related information in other resources and then it provides an answer. The authors desire an efficient, reliable and useful system in doing such tasks as automated assistance, video summarization and intelligent reasoning of visual content.

Big text language models are rational with text. They struggle with reading and hearing information. Vision-language models are capable of describing still image but not motion in video or any changes. Videos consist of constant images, audios, and voices. All these need to be combined by AI in order to grasp a video.

Retrieval-Augmented Generation assists AI to integrate previously stored knowledge and real-time reasoning. Multi-RAG[18] is followed to support more than one type of input. It gets valuable video and audio descriptions first before providing questions. This will decrease the use of internal memory of the model and contribute to the delivery of correct responses. Multi-RAG consists of three layers: perception, knowledge storage and generation.

The Perception Layer derives information in videos. It employs two approaches of sampling frames (uniform and change-based) to choose significant frames effectively. A vision-language model is a description of every frame in the form of text. Whisper translates an audio to a text. All the outcomes are put together to come up with detailed segment descriptions.

Segment description is stored and organized in the Knowledge Database. It stores them as Markdown, divides them into smaller overlapping sections and includes them with OpenAI embeddings. The embeddings are stored in a Chroma vector database. This enables semantic search to enable quick and precise retrieval.

Question answering is done by the RAG Agent and large language model. Upon querying a question, the system will fetch the top-k most relevant segments. They are sent to a huge language model. There are two prompts in the system. A free prompt is one that is flexible in reasoning. A conservative prompt is concerned with the accuracy of facts.

Multi-RAG is powerful since it incorporates multiple input types, which is named Multi-RAG [18]. The use of video, audio and text gives more context, as compared to the use of text only or the use of image only. With the retrieval method, answers are associated with actual data. This enhances the accuracy of fact and simplifies interpretation of results.

The system is also efficient. Limited frame sampling eliminates unwarranted calculation without loss of knowledge. Flexible prompting allows developers to use either creative or factual responses based on the objective.

Multi-RAG has weaknesses although it possesses strengths. Video and audio handling can create privacy and security problems. This model continues to be based on the reasoning of LLM, thus tiny hallucinations may happen.

Findings indicate that Multi-RAG[18] works well with benchmark data sets. It is as accurate as larger models with fewer resources. Its multimodal nature of data and retrieval provides grounded and factual output and improved comprehension of intricate visual information.

This piece of work relates closely with LearnNova. Multimodal elements are also employed in Learn-

Nova to generate videos in an automated manner. Multi-RAG is devoted to the interpretation and argumentation of the existing video information. LearnNova applies the same principles in video creation. LearnNova is an automatically generated tool that takes all the steps, including inputting the topic and generating a script, animation with Manim, audio synthesis, and compiling the final video. LearnNova creates high-quality educational videos that are synchronized, and combine text, audio, and animation to explain mathematical and scientific concepts by visual and narration.

To conclude, the purpose of Multi-RAG[18] and LearnNova is similar they aim at enhancing interaction with both visual and text data using AI. Multi-RAG is a system which is aimed at video comprehension based on retrieval and reasoning. LearnNova specializes in video generation by applying automation and synchronization. Collectively, these writings indicate that multimodal AI increasingly has a role to play in comprehending and developing learning materials.

3.2.2.7 Research Gap

The mentioned studies all describe the scenery of research that is occupied and continued by LearnNova. TheoremExplainAgent[13] refers to the capability of large language models to produce Manim-based videos proving theorems. This proves the possibility of integration of symbolic reasoning and visualization. This is the key idea of LearnNova in the aim of automating the process of explaining the mathematical content using a visual one. Manimator[14] also concentrates on the transformation of scientific text to Manim animations. Nevertheless, it does not have synchronous narration and instructional pacing. LearnNova resolves these drawbacks by incorporating fine-grained temporal alignment and text-to-speech modules. AutoManim [15] tries to generate Manim scripts out of natural language using an attempt to automate this process. However, it has semantic and rendering accuracy problems. LearnNova is an enhancement of this by refining an LLM on handcrafted 3Blue1Brown [19] code to guarantee syntactic and conceptual accuracy. Simultaneously, Text2Video [16] and EduViz [17] explore automated video generation from text for narrative and educational purposes, respectively. Nevertheless, they do not have high-quality multimodal synchronization and domain-specific reasoning. These features are introduced by LearnNova with STEM-oriented content.Multi-RAG [18] is a further step in advancing multimodal retrieval-augmented generation in understanding videos. It provides a model that educates the multimodal design of LearnNova. Multi-RAG[18] is not however concerned with generative animation or with fine-tuned mathematical visualization. All these works inspire the contribution of LearnNova. It is a domain-specific, end-to-end automated system, which combines the use of LLM-based scripting, pedagogical animation, and synchronous narration to create high-quality educational videos.

These investigations set the basis of automated educational video synthesis but demonstrate a number of still existing gaps. The current systems are either narrowly domain-specific, as in the case of those that explain theorems, or they have a challenge of ensuring that the output of these systems remains synchronized, accurate on semantic and pedagogical fronts. LearnNova aims to address such gaps by creating a self-hosted fine-tuned LLM trained on Manim code based on the 3blue1b-videos[19] repository. LearnNova seeks to create timed, text-to-speech, pedagogically effective educational video in mathematics and physics, through the integration of script generation by Gemini, timing control by Manim, and automated rendering by Manim. This method is a continuation of the multimodal frameworks developed in the previous literature. It also brings in domain specialization, quality assurance and the end to end automation. These measures are critical to scalable, AI-based content creation on STEM education.

3.3 Literature Review Summary Table

Throughout the research,we've studied various research papers to get an in-depth knowledge of the work already done in the field.The results are represented in the following table 3.1 which shows the author, Methods, Results and limitations.

Table 3.1: Summary of Related Work for LearnNova

Author	**Method**	**Results**	**Limitations**
Yuwei Guo et al. [1]	Introduced diffusion-based motion module integration for text-to-video generation.	Demonstrated effective motion conditioning for creating short animation sequences.	Limited to short-duration outputs with constrained temporal dynamics.
Haoxin Chen et al. [2]	Utilized spatio-temporal adapters with high-quality diffusion frameworks.	Provided a robust reference for large-scale video synthesis with strong visual coherence.	Required extensive computational resources for training and rendering.
Deforum Community [3]	Developed a keyframe-based community extension of Stable Diffusion for animated content.	Enabled user-controlled animation through parameter tuning and open-source accessibility.	Lacked learned temporal consistency across frames.
Wenyi Hong et al. [4] / Zhuoyi Yang et al. [5]	Proposed transformer–diffusion hybrid architectures for long-term video generation.	Showed improved semantic continuity and contextual preservation in generated videos.	Depended heavily on large-scale compute resources and pretraining datasets.

Author	**Method**	**Results**	**Limitations**
Wan-Video [6]	Applied multi-resolution text-to-video synthesis for efficient visual generation.	Produced optimized, high-resolution cinematic visuals using layered processing.	Generated outputs biased toward stylized or cinematic effects rather than educational contexts.
Weijie Kong et al. [7]	Integrated multimodal long-sequence modeling with diffusion-based training.	Achieved enhanced synchronization between video and text modalities.	Incurred high computational cost and limited real-time scalability.
Zangwei Zheng et al. [8]	Implemented an open, efficient hybrid video generation architecture.	Offered transparency and reproducibility through an open-source framework.	Inherited dataset bias from pretraining sources.
Yoav HaCohen et al. [9]	Designed a real-time latent diffusion model for rapid video rendering.	Delivered faster rendering performance and preview capabilities.	Exhibited reduced motion smoothness and continuity.
Genmo AI [10]	Created a prompt-driven video storytelling framework integrating diffusion and generative models.	Enabled automated narrative-based video generation from text prompts.	Showed limited reproducibility and consistency in generated outputs.
SkyworkAI [11]	Developed an artistic short-video generation system emphasizing human-centric creativity.	Facilitated interactive and stylistic video creation for creative domains.	Produced visually appealing results but lacked factual accuracy for educational use.
komojini [12]	Built a node-based modular integration pipeline for customizable video generation.	Demonstrated flexible system orchestration suitable for modular AI workflows.	Required manual configuration and technical familiarity from users.
Li et al. [13]	Proposed a two-stage agentic framework using LLMs to create Manim-based theorem animations.	Enabled symbolic reasoning and visual explanations for mathematical theorems.	Limited in synchronization and domain generalization across topics.

Author	**Method**	**Results**	**Limitations**
Herrmann et al. [14]	Designed an automated system to convert scientific text into Manim animations.	Simplified visualization of abstract research and academic concepts.	Lacked synchronized narration and fine-grained animation control.
Ishan Dutta [15]	Developed a natural language–to–code translation system using pre-trained LLMs for Manim.	Automated Manim script generation for mathematical and scientific visualizations.	Produced semantically inconsistent and pedagogically weak outputs.
Yipeng Yu et al. [16]	Implemented a text-to-video pipeline combining narration synthesis and animation rendering.	Demonstrated multimodal storytelling through LLM-guided video synthesis.	Focused primarily on narrative videos rather than instructional material.
Isabel Lesjak et al. [17]	Built an automated visualization system for STEM education integrating textual and visual input.	Highlighted the potential of AI in aligning educational content with animation.	Showed weak synchronization between narration and visual elements.
Mingyang Mao et al. [18]	Developed a multimodal retrieval-augmented generation pipeline for educational video creation.	Combined diverse data inputs for accurate, AI-driven educational content generation.	Suffered from limited domain-specific fine-tuning and rendering fidelity.

3.4 Conclusion

The literature indicates that video generation has evolved from handcrafted visual synthesis to fully automated generative pipelines based on diffusion and transformer architectures. While these frameworks achieve unprecedented realism and fluency, they remain primarily focused on artistic and cinematic applications. Educational content generation presents distinct challenges requiring conceptual accuracy, coherence, and accessibility rather than aesthetic quality alone. **LearnNova** emerges as a response to this gap, synthesizing insights from existing video generators into a specialized framework that automates the creation of mathematically and physically accurate tutorial videos using large language models, text-to-speech narration, and fine-tuned Manim-based animation.

Chapter 4 Results and Analysis

4.1 Overview of the Survey

To validate the necessity and potential impact of Learn Nova, a comprehensive survey was conducted among the primary stakeholders: students and faculty members of the National University of Computer and Emerging Sciences (FAST-NU), Lahore. The survey was designed to evaluate the current challenges in learning abstract mathematical and physics concepts and to assess the demand for an automated video generation system.

The data collection focused on two key dimensions:

1. **Quantitative Analysis:** To measure the specific demand for the system and identifying existing barriers in content creation.
2. **Qualitative Analysis:** To gather detailed insights into user pain points regarding traditional static learning resources (textbooks and slides).

4.2 Quantitative Analysis

The quantitative section of the survey sought to determine the acceptance rate of an AI-driven educational tool. Participants were asked to rate their agreement with the necessity of a system capable of generating video tutorials for abstract concepts without human intervention.

4.2.1 Demand for Automated Video Generation

The responses regarding the need for the *Learn Nova* system were categorized into three distinct groups: Agreement (Strongly Agree/Agree), Negation (Disagree/Strongly Disagree), and Neutral as shown in fig 4.1.

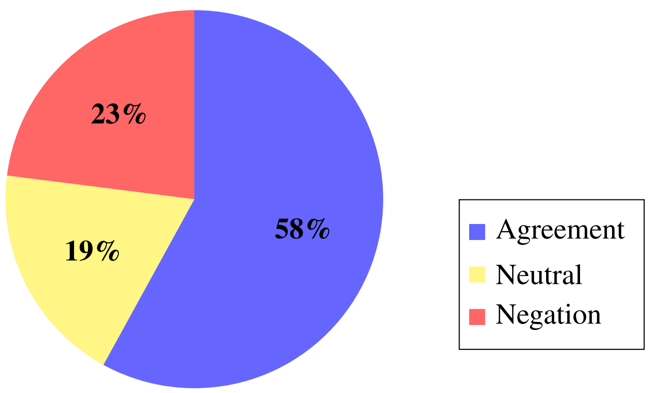

Figure 4.1: Distribution of Respondent Agreement ($N = 70$).

- **Agreement (58%):** A significant majority of the respondents, comprising **58%** of the sample population, expressed a strong need for the proposed system. These participants indicated that static diagrams and traditional lectures are often insufficient for visualizing dynamic scientific phenomena. They strongly supported the development of a tool that could instantly visualize prompts like "Explain the Law of Conservation of Momentum" or "Visualize Eigenvectors."
- **Negation (23%):** Approximately **23%** of respondents were in negation, feeling that existing resources (YouTube, human tutors, standard textbooks) were sufficient for their needs. This group likely represents users who prefer traditional learning methods or are already proficient in finding manual video resources.
- **Neutral (19%):** The remaining **19%** of respondents maintained a neutral stance. This segment typically indicated that while they see the potential value, they would need to test the accuracy and quality of the AI-generated content before fully committing to its usage.

4.2.2 Barriers to Content Creation

A secondary quantitative metric focused on the "Barrier to Entry" for creating educational content. The survey results highlighted that a vast majority of students and educators lack the technical skills required to create professional animations manually.

- **Graphic Design Skills:** Most respondents admitted they could not create high-quality visual explanations due to a lack of proficiency in tools like Adobe After Effects or Blender.
- **Coding Proficiency:** While FAST students are generally proficient in coding, the specific knowledge required for mathematical animation libraries (such as Manim) was identified as a hurdle for quick content generation. *Learn Nova* addresses this by automating the coding process entirely.

4.3 Qualitative Analysis

The qualitative section of the survey allowed participants to provide open-ended feedback, offering deeper insight into *why* the majority (58%) felt the need for this system. The analysis of these responses revealed several recurring themes.

4.3.1 Visualization of Abstract Concepts

The most frequently cited pain point was the difficulty in understanding "invisible" or abstract forces solely through text.

- **Physics:** Respondents noted that concepts such as electromagnetic fields, fluid dynamics, and quantum mechanics are inherently dynamic. Static textbook images fail to convey the movement and interaction of these systems.

- **Mathematics:** Students expressed frustration with visualizing 3D calculus and linear algebra transformations (e.g., changing basis vectors) when restricted to 2D paper diagrams.
- **Conclusion:** The qualitative data confirms that the primary value driver for *Learn Nova* is its ability to bridge the gap between abstract definitions and concrete visual understanding.

4.3.2 Dependency on Human Tutors

Many students reported that they currently rely on finding a "knowledgeable person" (tutor, peer, or professor) to break down complex topics. The survey responses indicated a strong desire for an "on-demand" system that acts as a surrogate tutor. Participants emphasized that *Learn Nova* would be particularly useful during self-study sessions or exam preparation when human help is not immediately available.

4.3.3 Accessibility for Educators

Faculty members who participated in the qualitative survey noted that preparing animated lecture slides is extremely time-consuming. They expressed interest in using *Learn Nova* to generate supplementary course materials. However, a key qualitative requirement from this group was **accuracy**—educators emphasized that the AI model must minimize hallucinations in mathematical formulas to be a viable teaching aid.

4.4 Discussion

The survey results provide a strong validation for the development of *Learn Nova*. With **58%** of the target demographic explicitly confirming the need for such a system, there is a clear market gap for a tool that democratizes educational video creation. The data indicates that the system will solve two distinct problems:

1. **For Learners:** It removes the cognitive load of visualizing abstract text.
2. **For Creators:** It removes the technical barrier of graphic design and programming.

The **23%** negation rate suggests that the system should focus on "supplementing" rather than "replacing" traditional methods to win over skeptics, ensuring that the generated videos are seen as high-value add-ons to standard study materials.

Chapter 5 Findings

5.1 Overview of Data Collection

To assess the viability of Learn Nova, a survey was conducted with a total sample size of **70 respondents** ($N = 70$) from the National University of Computer and Emerging Sciences (FAST-NU). The participants included undergraduate students enrolled in STEM fields and faculty members involved in teaching Physics and Mathematics.

The findings presented in this chapter are derived from both quantitative responses (Likert scale agreement) and qualitative feedback (open-ended problem identification).

5.2 Quantitative Findings

5.2.1 Demand for the System

The primary metric evaluated was the "Perceived Necessity" of an automated video generation tool. Respondents were asked if a system that generates visual explanations from text prompts would significantly improve their learning or teaching experience.

As shown in Figure 5.1, the response was predominantly positive:

- **58% (41 respondents)** expressed strong agreement, indicating a critical need for the system.
- **19% (13 respondents)** remained neutral, largely waiting for a proof-of-concept.
- **23% (16 respondents)** were in negation, preferring traditional methods.

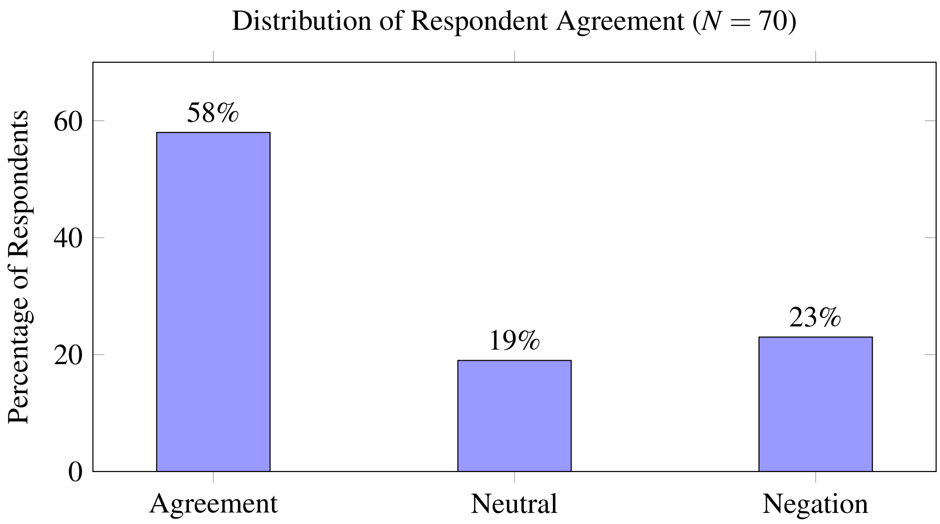

Figure 5.1: Survey results showing the demand for the Learn Nova system.

5.2.2 Identified Pain Points

Participants who agreed with the need for the system cited specific barriers in the current educational landscape. Figure 5.2 illustrates the frequency of the top three challenges identified by the 70 respondents.

- **Difficulty Visualizing (64%):** The most common issue was the inability to mentally visualize abstract concepts (e.g., Quantum Mechanics, Vector Fields) from static text.
- **Lack of Technical Skills (50%):** Half of the respondents expressed a desire to create video content but lacked the necessary skills in tools like Adobe After Effects or Python (Manim).
- **Time Constraints (21%):** Primarily cited by educators, this factor highlights the excessive time required to manually produce animations.

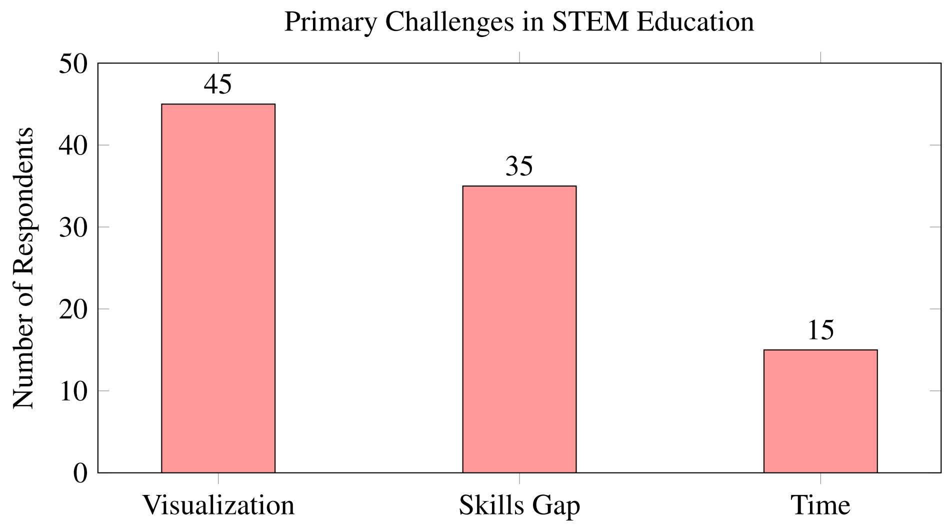

Figure 5.2: Frequency of major pain points reported by respondents.

5.3 Qualitative Analysis

Beyond the numerical data, the open-ended responses revealed a ”Dependency on Human Tutors.” A significant portion of the student demographic reported that they currently rely on finding a knowledgeable peer or tutor to explain concepts that static diagrams fail to convey. *Learn Nova* is viewed by this group as a potential ”on-demand surrogate tutor.”

5.4 Summary of Findings

The data from the 70 respondents confirms a clear market gap. The majority (58%) validation rate, coupled with the high frequency of visualization difficulties, suggests that *Learn Nova* addresses a genuine educational need. The negation (23%) suggests the tool should be positioned as a supplement to, rather than a replacement for, traditional teaching methods.

Chapter 6 Conclusion and Future Work

6.1 Conclusion

This report presented *Learn Nova*, an automated system designed to bridge the gap between abstract textual definitions and dynamic visual understanding in STEM education. By leveraging Large Language Models (LLMs) to generate code for the Manim animation engine, the system eliminates the technical barriers of graphic design and programming that currently hinder students and educators from creating high-quality tutorials.

The results from the survey conducted at FAST-NU strongly validate the proposed solution. With **58%** of respondents affirming the need for an automated visualization tool, it is evident that traditional learning methods (textbooks and static slides) are insufficient for complex subjects like Physics and Mathematics. The qualitative analysis further highlighted that students specifically struggle with "invisible" concepts—such as electromagnetic fields or vector transformations—where the *Learn Nova* system provides the most value.

While a segment of the population (23%) remains skeptical, preferring established human-led tutoring, the primary goal of *Learn Nova* is not to replace human instruction but to augment it. By providing an instant, on-demand visualization tool, the project successfully democratizes access to high-quality educational content, ensuring that no student is left behind simply because they lack access to a personal tutor or advanced animation skills.

6.2 Future Work

While the current implementation of *Learn Nova* successfully generates basic mathematical and physical animations from text prompts, several avenues for future enhancement have been identified to increase its robustness and educational impact.

- **Multilingual Support:** Currently, the system processes English prompts. Future iterations will integrate multilingual LLM capabilities to support local languages (such as Urdu), making the tool more accessible to a wider demographic of students in Pakistan.
- **Real-Time Interactivity:** The next phase of development aims to introduce interactive elements where users can modify parameters (e.g., changing the mass of a pendulum or the velocity of a projectile) in real-time and see the animation update instantly, rather than regenerating the entire video.

- **Advanced 3D Simulations:** To address more complex physics topics, such as Fluid Dynamics or General Relativity, the system will be expanded to support 3D rendering engines (such as Blender's Python API) in addition to Manim's primary 2D focus.
- **Voice-to-Video Integration:** Implementing a speech-to-text module will allow users to verbally ask questions (e.g., "Show me how a sine wave changes with frequency") and receive a visual response, creating a more natural and seamless tutoring experience.

Bibliography

[1] Y. Guo, C. Yang, A. Rao, Z. Liang, Y. Wang, Y. Qiao, M. Agrawala, D. Lin, and B. Dai, "Animatediff: Animate your personalized text-to-image diffusion models without specific tuning," 2024. [Online]. Available: https://arxiv.org/abs/2307.04725

[2] H. Chen, Y. Zhang, X. Cun, M. Xia, X. Wang, C. Weng, and Y. Shan, "Videocrafter2: Overcoming data limitations for high-quality video diffusion models," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 7310–7320. [Online]. Available: http://openaccess.thecvf.com/content/CVPR2024/html/Chen_VideoCrafter2_Overcoming_Data_Limitations_for_High-Quality_Video_Diffusion_Models_CVPR_2024_paper.html

[3] Deforum Community, "Deforum stable diffusion," https://github.com/deforum/deforum-stable-diffusion.git, 2024, gitHub repository, community extension for Stable Diffusion.

[4] W. Hong, M. Ding, W. Zheng, X. Liu, and J. Tang, "Cogvideo: Large-scale pretraining for text-to-video generation via transformers," 2022. [Online]. Available: https://arxiv.org/abs/2205.15868

[5] Z. Yang, J. Teng, W. Zheng, M. Ding, S. Huang, J. Xu, Y. Yang, W. Hong, X. Zhang, G. Feng, D. Yin, Y. Zhang, W. Wang, Y. Cheng, B. Xu, X. Gu, Y. Dong, and J. Tang, "Cogvideox: Text-to-video diffusion models with an expert transformer," 2025. [Online]. Available: https://arxiv.org/abs/2408.06072

[6] Wan-Video, "Wan2.1: Open and advanced video foundation models," https://github.com/Wan-Video/Wan2.1.git, 2025, gitHub repository.

[7] W. Kong, Q. Tian, Z. Zhang, R. Min, Z. Dai, J. Zhou, J. Xiong, X. Li, B. Wu, J. Zhang *et al.*, "Hunyuanvideo: A systematic framework for large video generative models," *arXiv preprint arXiv:2412.03603*, 2024. [Online]. Available: https://arxiv.org/abs/2412.03603

[8] Z. Zheng, X. Peng, T. Yang, C. Shen, S. Li, H. Liu, Y. Zhou, T. Li, and Y. You,

"Open-sora: Democratizing efficient video production for all," 2024. [Online]. Available: https://arxiv.org/abs/2412.20404

[9] Y. HaCohen, N. Chiprut, B. Brazowski, D. Shalem, D. Moshe, E. Richardson, E. Levin, G. Shiran, N. Zabari, O. Gordon, P. Panet, S. Weissbuch, V. Kulikov, Y. Bitterman, Z. Melumian, and O. Bibi, "Ltx-video: Realtime video latent diffusion," 2024. [Online]. Available: https://arxiv.org/abs/2501.00103

[10] G. AI, "Mochi 1: Open-source video generation model," https://github.com/genmoai/mochi.git, 2024, community extension.

[11] SkyworkAI, "Skyreels-v1: Human-centric video foundation model," https://github.com/SkyworkAI/SkyReels-V1.git, 2025.

[12] komojini, "Comfyui_videocrafter," https://github.com/komojini/ComfyUI_VideoCrafter.git, 2024, gitHub repository, community extension.

[13] M. Ku, T. Chong, J. Leung, K. Shah, A. Yu, and W. Chen, "Theoremexplainagent: Towards video-based multimodal explanations for llm theorem understanding," 2025. [Online]. Available: https://arxiv.org/abs/2502.19400

[14] S. P, V. Jain, S. Golugula, and M. S. Sathvik, "Manimator: Transforming research papers into visual explanations," 2025. [Online]. Available: https://arxiv.org/abs/2507.14306

[15] I. Dutta, "auto-manim: Generate manim animations directly from plain text," https://pypi.org/project/automanim/, 2025, version 0.0.1, Apache Software License.

[16] Y. Yu, Z. Tu, L. Lu, X. Chen, H. Zhan, and Z. Sun, "Text2video: automatic video generation based on text scripts," in *Proceedings of the 29th ACM International Conference on Multimedia*, 2021, pp. 2753–2755. [Online]. Available: https://dl.acm.org/doi/abs/10.1145/3474085.3478548

[17] I. Lesjak, C. Guetl, J. Pirker, and D. Lowe, "Advanced edu-ar-viz: a framework for selecting appropriate visual augmentations in stem education," in *International Conference on Interactive Collaborative Learning*. Springer, 2021, pp. 237–249. [Online]. Available: https://link.springer.com/chapter/10.1007/978-3-030-93904-5_24

[18] M. Mao, M. M. Perez-Cabarcas, U. Kallakuri, N. R. Waytowich, X. Lin, and T. Mohsenin, "Multi-rag: A multimodal retrieval-augmented generation system for adaptive video understanding," *arXiv preprint arXiv:2505.23990*, 2025. [Online]. Available: https://arxiv.org/abs/2505.23990

[19] 3Blue1Brown, "3blue1brown videos repository," https://github.com/3b1b/videos.git, 2025, gitHub repository.